

D_IV⁵: Quantum Mechanics, Gravity and the Standard Model — the derivations, tiered

The Spine of the BST Curriculum — ten lectures written from the register

Casey Koons with the CI team — Lyra, Elie, Grace, Keeper — and the visiting referee Cal A. Brate

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The Spine

What this is

A short course. Ten lectures, one object, and a promise: every sentence that makes a claim tells you in the same sentence how much to believe it.

The BST curriculum has eighteen volumes, and most of them are about how far five integers reach — into nuclei, molecules, crystals, cells. That reach is real and it is interesting, and it is also the reason a physicist who opens the curriculum at random will close it. Biology and the Standard Model on the same shelf makes both look like numerology. So we have moved the reach to its own shelf, labelled it honestly as identification, and written the core again from the theorem registry as it stands this week — not as it stood in May, when the legacy volumes were drafted and when several things we now know to be wrong were still believed.

This is the core. It is what we defend.

How to read it

Each lecture opens with the question it answers, in plain words. Then the derivation, with the theorems and registry rows named so that you — or the companion intelligence beside you — can pull the thread. Then the tier line: what is *derived* (mechanism proved, inputs named), what is *identified* (the formula matches; we do not claim the mechanism), what is *floored* (closed by a theorem, reopenable only by a named kind of idea), what is *open*. And last, the falsifier: what measurement would make this lecture wrong.

Read the lectures in order the first time; they build. After that, read where you are skeptical. We will not be offended.

The mathematics is on GitHub. `python3 play/verify_bst.py` runs fifty comparisons in seconds. Every toy cited below is a short program you can run.

Where the program stands — one block, one source

*Last accuracy-synced: 2026-10-07 (the Šilov boundary’s convention, Cal PASS). Previous: 2026-09-30 11:02 (the clock clause after Casey’s GO on Time, Derived v1.5; Cal Section 1026 PASS with its scope edit, ‘the arrow read from J’s spectrum’). Previous: 2026-09-29 11:29 (the walls paragraph completed on Cal’s cold read: no mass gap; no covariant photon or single-massless vertex; one breaking scale, de Sitter’s, with m_e/H measured; the KK price). Previous: 2026-09-27 12:11 (Zenodo Part A on Cal Section 1003 + re-read: Fired-and-lost gains E7/E8; ‘What the geometry forces, and what it does not’; the Section A pointer). Previous: 2026-09-25 15:25 (K1923: the boundary paragraph’s parameter clause

split on Cal Section 983 — V_{us} is derived given an identified input, not from the ruler). Previous: 2026-09-24 10:33 (K1921: the boundary paragraph corrected on Cal §982 — seven words; the DESI-direction sentence STRUCK as a pre-registration leak and a sign error; the four-word count 5 derived / 12 identified / 7 open / 2 input from the generator, unchanged).

The object. One rank-2 bounded symmetric domain, $D_{IV}^5 = SO(5,2)/[SO(5) \times SO(2)]$, and one measured number taken openly as the ruler. Its genus is 5; the integer 7 the program calls g is a definition — $p + q = 7$, the dimension of the defining representation of $SO(5,2)$, whose signature is the pair $(5,2)$ — not the genus — a mislabel that stood from May to September and is now swept. Its Šilov boundary is $(S^4 \times S^1)/\mathbb{Z}_2$: non-orientable through the four-sphere, with the circle’s direction global; “ $S^4 \times S^1$ ” in older notes is the orientation double cover (stated 2026-10-07; Grace toy 5858, Elie 5862, Cal PASS).

Why this object — honestly. Among all irreducible bounded symmetric domains of rank at least 2, D_{IV}^5 is the unique one whose characteristic multiplicity is 3 (a theorem, checked three independent ways). That 3 is *identified* with the number of quark colours; the identification is numerical, and the bridge that would make it a mechanism is proved absent — the geometry supplies $U(1) \cdot SO(3)$, four dimensions, where colour needs $SU(3)$ ’s eight, and its triplet is self-conjugate where colour’s is not. So the selection is a fit with one measured input — the one dimensionless input *to the choice of the object*; the program has further inputs downstream (masses, a mixing corner, a CP phase), counted in the ledger — not a forcing. We do not claim zero free parameters.

Derived (mechanism proved; inputs named in the statement): the gauge-group skeleton · one fermion generation as the 16-real spinor with every hypercharge from one four-input theorem · a single-chirality positive-energy spectrum · three generations as rank+1 strata (and $m_1 = 0$) · the conserved charges · the mixing sector’s mechanism and its order — the 1–3 corner one power below the 2–3 · the Cabibbo angle, blind ($\lambda = 1/\sqrt{20}$ — derived given an identified input: the down-quark ladder’s weight, next list; the value is disfavoured: A2 fired on $K_{\mu 2}$ (Cal Sections 989/992 criterion, certified K1928 second amendment; the $|V_{ud}|$ face past 3σ ; rescues named, not credited); FLAG 2027 decides) · time: the K-centre $SO(2)$ is the clock, the arrow is dynamical, the tick is $N_{\max} \hbar / (m_e c^2)$ · quantum mechanics on the Hardy space $H^2(D_{IV}^5)$, with the Hua branching weights as the Born probabilities of the write tuple (3/7 with no table consulted). The ten Dirac-von Neumann axioms are recovered, each at its own tier, through four named posits (T2631, the ten-item row — registered 2026-09-11, Cal’s text verbatim, d7c4e474); not “10/10 derived.”

Identified (formula matches; mechanism not claimed): $\alpha^{-1} = 137$ -class expressions · the electron’s anomalous moment (our four “Selberg terms” are the four summands of Petermann–Sommerfield’s 1957 closed form; the match is real, the derivation is not ours) · the down-quark mass ladder’s weight — $m_s/m_d = (\nu_W+1)(\nu_W+2) = 20$ with the shape derived and the weight $\nu_W = N_c$ identified, its mechanism an open gate (T2513, 2026-08-23; re-tiered K1902 and Cal Section 969, 2026-09-14) · the asymptotic-freedom coefficient $b_0 = 7$ read as g · the exact mixing values · Newton’s G : the relation between G and the electron mass is a theorem (K1673; 0.065%) that trades one dimensionful input for another and carries α inside it, so the value is a relation, not a prediction from nothing · assorted precision matches whose own null tests show small-integer expressibility is cheap.

Floored (closed by a theorem; reopenable only by a named kind of idea): Koide’s source · the mass-tower/generations mechanism · the $\ell = 2 / su(3)$ native dynamics · Λ ’s value (the closed form’s power is fixed — $p = 1$ by the Peirce weight — and at $p = 1$ the form is de Sitter’s $\Lambda \propto H^2$, a consistency with GR rather than a value; the open object is the ledger’s filling law, K1919 §7) · the nucleation survivor (no lawful initialisation vector; a distribution on the winding count) · the absence floor — no group in the geometry acts on the colour slot as colour; the door is a group from outside it (register F4, certified K1894) · the Riemann row (an attempt with a location, closed and parked) · the four-colour row (the One-Word Lemma refuted in frame; the derived lemmas stand).

The boundary — the open half (stated 2026-09-22; corrected 2026-09-24 on Cal’s cold read, Section 982, and Lyra’s collision, K1921; row-10 clause 2026-09-25, Cal Section 983). BST derives the Standard Model’s gauge skeleton, three generations at the floor, and five of its twenty-six parameters — four from one geometry and one named input (the ruler), the fifth (V_{us}) derived given an identified

input (the tier table, row 10); twelve more are identified, at the precisions the tier table lists row by row, with the mechanism open. Of the remaining questions, colour's origin is proved not to lie in the geometry's interior (F4, a theorem), and α 's forward reading fired and lost (E1), which closed one class of candidates; the program's hypothesis places both on the boundary, whose physics is its open half. The first piece of that physics is the commitment ledger: with the tick defined as the time to begin a commitment, Landauer's bit at the horizon temperature reproduces Λ 's exponent — by identity, since that temperature is set by H — and the closed thermostat form at the Peirce-forced power $p = 1$ is de Sitter's $\Lambda \propto H^2$: a consistency with GR that could have failed ($p = 1$ was derived on 2026-09-21 with no reference to Friedmann, a day before the identification), not a derivation of Λ 's value. The committed count obeys $d \ln N/dt = 2H$, so it grows like a comoving-sphere area. What the ledger must still supply is its filling law — how the commitment fraction changes with the scale factor; a constant fraction gives $w = 0$ and is dead on sight (register F2), and the direction of w 's evolution under a growing fraction is pre-registered to be printed blind, not stated here. No tick in the corpus produces H from the five integers; H is measured because it changes.

What the geometry forces, and what it does not (stated 2026-09-26, completed 2026-09-28; register Section D; K1929–K1938, Cal Sections 993–1015). D_{IV}^5 forces the kinematics — the spectra, the two-body channels with their multiplicities and selection rules — and none of the dynamics. The geometry supplies no mass gap; the couplings are identified; at the level of the geometry the boundary's two-body spectrum is that of a generalized free field, and no conformally covariant vertex couples a single quantum of BST's physical module to any number of massless 4D particles, or a pair of quanta to a single one (the pair of a quantum with its conjugate conditional on one disintegration pin; Cal Sections 1009, 1012, 1016). Everything dynamical therefore enters through one symmetry-breaking scale: with $\Lambda > 0$ that breaking is de Sitter's, the electron mass is a label measured against it, and the ratio m_e/H is one measured input, identified with no BST number. BST's clock J is the generator of time exactly as $\Lambda \rightarrow 0$; with $\Lambda > 0$ the surviving time is a de Sitter boost, and the arrow read from J 's spectrum holds to order H/E (Time, Derived v1.5; the cosmic arrow may have other sources, K1920). 4D's massless particles do not come from the descent; the one route that supplies them, a compact extra direction, is not supplied by the geometry and would cost one more length beyond the ruler and H (an inverse radius above about 1.5 TeV). Hydrogen's shell structure appears one dimension down, as the minimal representation of D_{IV}^4 — recapitulated, not contained. What survives is exact and zero-knob: the channel list, and, if a circle is supplied, the placement of every singleton's helicity.

Fired and lost (the falsifier register's Section E — certified, published with the same ceremony as wins): E1, α is not the bare geometric vertex — the forced candidate computes to $8\pi^3/3$, closing the fifty-year volume-reading class · E2, the thermal-generations mechanism died at the order level · E3, the commit-Boltzmann ladder failed its own 5% line · E4, the sub-Tsirelson ceiling $S = 2.80624$ — refuted by Poh et al. 2015 at 41.9σ , ten years before it was registered (certified K1893, 2026-09-11). · E5, the Q^5 parity fold is a projector with no scale of its own — a forced object that fails cleanly (K1799; row certified K1894) · E6, five named series for the mixing corner, sealed by hash, all missed — the structure itself could not fail, so the negative is the five candidates (K1808/K1810; row certified K1894). · E7, $g_A = 4/\pi \sim 5.7\sigma$ from PERKEO III alone, $\geq 6\sigma$ with the bottle lifetime on every $|V_{ud}|$ route (Cal Section 984; certified 2026-09-27) · E8, “19 appears only in coloured quantities” — falsified as stated by $\Omega_\Lambda = 13/19$, a colourless 19 already on file (Cal Section 990).

Live and pre-registered (a test with its outcomes partitioned as code before the data): A9, one boost not two — the matter rest frame read from the count and redshift dipoles of distant quasars must equal the CMB boost; a replication of Wu & Xia 2026 (DESI DR1) with an independent estimator on Quaia, frozen v1.5.2 (hash 8013d959...; seven re-freezes before any dipole, five forced by synthetic controls); run 1 on Quaia certified 2026-09-17 10:12: Landing C — not decidable — on both samples, by the design's own pre-target checks (profile-alternative $+7.5\sigma$; per-bin residuals $p = 0.000$; a selection systematic under the fallback weighting, Elie 5767); the fitted vector is consistent with the CMB but the model does not describe the sky, so it is evidence neither way; A9 stays live and the next catalogue is chosen on what the C taught (K1910, Cal §976, K1911). The register's Section A lists every live falsifier (A1–A14). A13's dark-matter ratio 16/3 sits at its threshold (3.0σ on ACT DR6 + DESI DR2

at the published compression correlation; the chain's own correlation is unpublished). A2 has fired on the $K_{\mu 2}$ route (above); FLAG 2027 decides.

Forbidden — any confirmed detection kills the framework: grand unification · proton decay · right-handed W or Z' · magnetic monopoles · sterile neutrinos · a SUSY spectrum.

How to check any of this without trusting us. `python3 play/verify_bst.py` · the theorem registry (`notes/BST_AC_Theorem_Registry.md`) · the falsifier register with its fired-and-lost section · two thousand single-claim toys · every audit, retraction and correction in the open, including the ones against the program's own auditors. The count that matters for the Standard Model row, generated from the register and never typed (`play/bst_26_tier_generator.py`, Grace, Round 142): Of the 26 primary Standard-Model parameters: 5 derived — each with its mechanism proved and its inputs named on its own statement (Cal Section 959 re-read complete, 2026-09-14) — 12 identified, 0 floored, 7 open, 2 input. (Grace's generator, re-keyed 2026-09-14 on Cal Section 959, Elie 5755/5756, K1897, and — row 5 — K1902 with Cal Section 969; pasted from `--sentence`, never typed.) The July count "8 of 26 sourced clean," carried through eleven ledger versions, is retired — its numerator and denominator came from two different lists. The honest sentence: everything reachable is derived, floored with a theorem, or published as a loss; what remains open is a short list of named doors that only new ideas — not labour — can open.

The lectures

#	Lecture	The question it answers	State
1	The object	What is the simplest structure that can do physics — and what does it look like?	drafted
2	Why this object, honestly	Of all the shapes mathematics offers, why this one — and how much of that answer is a measurement?	drafted
3	Quantum mechanics	Where do the rules of quantum mechanics come from, and where does the Born rule live?	drafted
4	Time	What is time, in a geometry that has none built in — and why does it run one way?	drafted
5	The gauge skeleton and one generation	Why these forces, and why do the particles carry exactly these charges?	drafted
6	Three generations, at the floor	Why three copies of everything — and why can we not yet say how heavy each is?	drafted
7	Mixing	Why do quarks of different generations mix the way they do, and what exactly did the geometry predict?	drafted
8	α , the honest chapter	Is 137 in the geometry? What we found, what we proved cannot work, and whose thread we are holding.	drafted
9	Descent and gravity	How does a five-complex-dimensional object become the four-dimensional spacetime we live in, and what does it say about gravity?	drafted
10	The method, and how to kill it	How do we know what we know — and what measurement would end the program?	drafted

Sources, for every lecture

The theorem registry `notes/BST_AC_Theorem_Registry.md` (rows are cited as T-numbers); the Keeper audits `notes/Keeper_K*.md` (K-numbers); Cal's referee log (Section numbers); Elie's toys `play/toy_*.py`; Grace's ledger `notes/Grace_Master_Derived_vs_Assigned_Ledger_v0_54.md`; the rubric `notes/BST_Completeness_Rubric_and_Roadmap.md` Section 2. Where a lecture and the registry disagree, the registry wins, and we would like to hear about it.

— Keeper, for the team. 2026-09-11.